Assignment (19/03/2026)

Smart Resume Analyzer with Skill Gap Detection and Career Roadmap

Problem Statement

Many students apply for jobs without knowing whether their resume matches industry requirements. They lack clarity on missing skills, resulting in repeated rejections and poor career direction.

Objective

To build an AI-powered system that analyzes a resume, compares it with job descriptions, and identifies skill gaps while suggesting a personalized learning roadmap.

Dataset

- Sample resumes (Kaggle Resume Dataset)
- Job descriptions (LinkedIn / Naukri scraped data)
- Skills dataset (predefined skill taxonomy like Python, ML, SQL, etc.)

Algorithms / Techniques

- **TF-IDF / BERT** → Resume & job description similarity
- **Named Entity Recognition (NER)** → Extract skills from resume
- **Cosine Similarity** → Match resume vs job role
- **Rule-based mapping** → Identify missing skills
- **Recommendation logic** → Suggest courses/resources

Methodology

1. User uploads resume (PDF/Text)
2. Extract text using NLP
3. Identify skills using NER
4. Compare with selected job role
5. Detect missing skills (skill gap)
6. Generate personalized roadmap
   (e.g., “Learn SQL → Build Project → Apply”)

Expected Output

- Extracted skills from resume
- Match percentage with job role
- Missing skills list

- Personalized learning roadmap
- Suggested projects/courses

Tools & Technologies

Python, NLP (spaCy / Hugging Face), Scikit-learn, Streamlit, PDF Parser

Novelty / Innovation

Unlike basic resume checkers, this system not only analyzes resumes but also provides actionable career guidance, helping students move from rejection → selection.

Future Scope

- Integration with LinkedIn
- Real-time job recommendations
- AI mock interview system